

Lecture 9: Sept 8

Last time

- Independence

Today

- Random variables

Example Let the sample space S consist of the $3!$ permutations of the letters a , b , and c along with the three triples of each letter. Thus,

$$S = \left\{ \begin{array}{ccc} aaa & bbb & ccc \\ abc & bca & cba \\ acb & bac & cab \end{array} \right\}.$$

Furthermore, let each element of S have probability $\frac{1}{9}$. Define

$$A_i = \{i^{th} \text{ place in the triple is occupied by } a\}.$$

What are the values for $\Pr(A_i)$, $i = 1, 2, 3$? Are they pairwise independent?

solution

Definition* A collection of events $A_1, \dots, A_n$ are *mutually independent* if for any subcollection $A_{i_1}, \dots, A_{i_k}$, we have

$$\Pr(\cap_{j=1}^k A_{i_j}) = \prod_{j=1}^k \Pr(A_{i_j}).$$

Random Variables

In many experiments, it is easier to deal with a summary variable than with the original probability structure.

Definition A *random variable* (r.v.) is a function from a sample space S into the real numbers.

Example In some experiments random variables are implicitly used

Examples of random variables

Experiment	Random variable
Toss two dice	X = sum of numbers
Toss a coin 25 times	X = number of heads in 25 tosses
Apply different amounts of fertilizer to corn plants	X = yield / acre

In defining a random variable, we have also defined a new sample space (the range of the random variable).

Induced probability function Suppose we have a sample space $S = \{s_1, s_2, \dots, s_n\}$ with a probability function $\Pr$ defined on the original sample space. We define a random variable X with range $\mathcal{X} = \{x_1, \dots, x_m\}$. We can define a probability function $\Pr_X$ on $\mathcal{X}$ in the following way. We will observe $X = x_i$ if and only if the outcome of the random experiment is an $s_j \in S$ such that $X(s_j) = x_i$. Therefore,

$$\Pr_X(X = x_i) = \Pr(\{s_j \in S : X(s_j) = x_i\}),$$

defines an *induced* probability function on $\mathcal{X}$, defined in terms of the original function $\Pr$.

We will write $\Pr(X = x_i)$ rather than $\Pr_X(X = x_i)$ for simplicity. Note on notation: random variables will always be denoted with uppercase letters and the realized values of the variable (or its range) will be denoted by the corresponding lowercase letters.

Example Consider the experiment of tossing a fair coin three times. Define the random variable X to be the number of heads obtained in the three tosses. A complete enumeration of the value of X for each point in the sample space is

s	HHH	HHT	HTH	THH	TTH	THT	HTT	TTT
$X(s)$	3	2	2	2	1	1	1	0

What is the range of X ? What is the induced probability function $\Pr_X$?

solution:

So far, we have seen finite S and finite S_X , and the definition of $\Pr_X$ is straightforward. If S_X is uncountable, we define the induced probability function, $\Pr_X$ for any set $A \subset S_X$,

$$\Pr_X(X \in A) = \Pr(\{s \in S : X(s) \in A\}).$$

This defines a legitimate probability function for which the Kolmogorov Axioms can be verified.

Distribution Functions

Distribution Functions are used to describe the behavior of a r.v.

Cumulative distribution function

Definition The *cumulative distribution function* or *cdf* of a random variable X , denoted by $F_X(x)$, is defined by

$$F_X(x) = \Pr_X(X \leq x), \text{ for all } x.$$

Definition The *survival function* of a random variable X , is defined by

$$S_X(x) = 1 - F_X(x) = \Pr_X(X > x).$$

Example Consider the experiment of tossing three fair coins, and let X = number of heads observed. The cdf of X is

$$F_X(x) = \begin{cases} 0 & \text{if } -\infty < x < 0 \\ \frac{1}{8} & \text{if } 0 \leq x < 1 \\ \frac{1}{2} & \text{if } 1 \leq x < 2 \\ \frac{7}{8} & \text{if } 2 \leq x < 3 \\ 1 & \text{if } 3 \leq x < \infty \end{cases}$$

Some properties of the cdf:

Let $F(x)$ be a cdf. Then

1. $0 \leq F(x) \leq 1$
2. $\lim_{x \rightarrow -\infty} F(x) = 0$
3. $\lim_{x \rightarrow \infty} F(x) = 1$
4. F is nondecreasing: if $a < b$, then $F(a) \leq F(b)$
5. F is right-continuous: $\lim_{x \downarrow b} F(x) = F(b)$, or $\lim_{x \rightarrow b^+} F(x) = F(b)$
6. $\Pr(a < X \leq B) = F(b) - F(a)$

Theorem The function $F(x)$ is a cdf if and only if the following three conditions hold:

1. $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$
2. F is nondecreasing: if $a < b$, then $F(a) \leq F(b)$
3. F is right-continuous: $\lim_{x \downarrow b} F(x) = F(b)$, or $\lim_{x \rightarrow b^+} F(x) = F(b)$

The cdf does not contain information about the original sample space.

Definition Two random variables X and Y are identically distributed if, for every Borel set $A \subset \mathbb{R}$, $\Pr(X \in A) = \Pr(Y \in A)$.

Example Toss a fair coin n times. The number of heads and the number of tails have the same distribution.

Theorem The following two statements are equivalent:

1. The random variables X and Y are *identically distributed*.
2. $F_X(x) = F_Y(x)$ for every x .

Types of Random Variables

Definition A random variable X can be

- *discrete*:
 - X takes on a finite or countably infinite number of values
 - $F_X(x)$ is step-wise constant
- *continuous*:
 - the range of X consists of subsets of the real line
 - $F_X(x)$ is continuous.
- *mixed*: $F_X(x)$ is piecewise continuous.

Example A random variable has cdf

$$F(x) = \begin{cases} 0 & x < 0 \\ x/2 & 0 \leq x < 1 \\ 2/3 & 1 \leq x < 2 \\ 11/12 & 2 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Is this a valid cdf? Is it a discrete random variable or continuous random variable or mixed?
solution: